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TECHNICAL MANUAL

PART	III	OVERHAUL AND RECONDITIONING INSTRUCTIONS
VOLUME	-	-
BOOK	1	

FIELD TELEPHONE SET IFT-1001

(Y1/5805-XXXXXX)

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8th Floor, Tower B, Windsor IT Park,
A-1, Sector-125, Noida (UP) - 201301
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RECORD OF CHANGES

Change Notice No.	Authority	Incorporated by "Name and Rank" in Block letters	Initials with date
(1)	(2)	(3)	(4)

ASSOCIATED HANDBOOKS

Technical Literature for **FIELD TELEPHONE SET IFT-1001** comprises of the followings:

- USER HANDBOOK
- TECHNICAL MANUAL

PART I		TECHNICAL INFORMATION
	Volume 1	Technical Description
	Volume 2	Drawings
PART II		MAINTENANCE
PART III		**OVERHAUL AND RECONDITIONING INSTRUCTIONS**
PART IV		MANUFACTURER'S PARTS LIST
	Volume 1	Parts List
	Volume 2	Illustrations

SAFETY WARNING

The voltages employed in this Equipment are sufficiently high to endanger human life.

POWER
MUST BE SWITCHED OFF
before servicing the equipment
&
GREAT CARE
taken when making internal adjustments etc.

FOR FIRST AID IN CASE OF ELECTRIC SHOCK SEE OVERLEAF

FIRST AID IN CASE OF ELECTRIC SHOCK

FIG. 1.

FIG. 2.

FIG. 3.

FIG. 4.

1.	SWITCH OFF. If this is not possible, PROTECT YOURSELF with dry insulating material and pull the victim clear of the conductor. DO NOT TOUCH THE VICTIM WITH YOUR BARE HANDS until he is clear of the conductor, but DO NOT WASTE TIME.
2.	(a) Place the victim in the supine position. (b) Keep the air passages clear by turning the head to one side, opening the patient's mouth and clearing it of water, saline, mucus or blood, a lot of which might have accumulated in the back of the throat.(Figure 1).
3.	If the jaw is rigid try to force the mouth open by pressure on the gum behind the last molar tooth of the lower jaw. When the upper air passages are thus cleared, tilt the head backward and force the jaw forwards from the angles of the jaw in front of the ears. This would prevent mechanical obstructions to the upper air passages.(Figure 2-3)
4.	(a) Then hold the chin up and forward with one hand and pinch the nostrils of the victim with the other.(Figure 4) (b) Take a very deep breath and apply your mouth to that of the victim and blow into his mouth, until the chest of the victim moves up indicating filling of the lungs. (NEVER ALLOW THE CHIN TO SAG).(Figure 4)
5.	When the chest has moved up, withdraw your mouth and allow the chest to slink back. REPEAT this process every three to four seconds until the victim begins to breathe again or until he is taken over by a medical attendant. This method can be continued in an ambulance during transit of the patient from the site of accident to the nearest medical centre.
HAVE SOMEONE ELSE AND SEND FOR A DOCTOR. KEEP PATIENT WARM AND LOOSEN HIS CLOTHING. **DO NOT GIVE LIQUIDS UNTIL THE PATIENT IS CONSCIOUS.**	

CAUTION/NOTES

Ringer output Voltage in this equipment is sufficiently high to endanger human life. Powers of external battery connections / internal batteries are to be removed before servicing the equipment.

CAUTION

Do not touch the exposed metal conductor of the field wire, when the phone is in Magneto position and Magneto Call button is pressed.

DEMOLITION OF MATERIAL TO PREVENT ENEMY USE

(A) General

The demolition procedure outlined below will be used to prevent the enemy from using or salvaging this equipment. **DEMOLITION OF THE EQUIPMENT WILL BE DONE ONLY ON ORDER OF THE COMMANDER**.

(B) Methods of Destruction

Use any or all the following methods to destroy the equipment.

(i) Smash - Smash the crystals, controls semiconductors, coil, switches, transformers and handset. Use sledges, hand axes, pickaxes, hammers crowbars and heavy tools.

(ii) Cut - Cut the cords, headsets and wiring. Use axes, hands axes or hatchets.

(iii) Burn - Burn the cords, resistors, capacitors, coils, wiring, technical manuals. Use gasoline, kerosene, oil, flamethrowers or incendiary.

(iv) Bend - Bend the panels, cabinet and chassis.

(v) Explosives - If explosives are necessary, use firearms, grenades or TNT.

(vi) Disposal - Bury or scatter the destroyed parts in slit trenches, fox holes or throw them into streams.

CHANGE NOTICE

TITLE OF PUBLICATION	CODE NO. AND DATE	PREVIOUS CHANGE	CHANGE NOTICE NO. AND DATE

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(H) By Hand – correction	
(P) By pasting enclosed matter	
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Figure 1 : Isometric view of Field Telephone Set IFT - 1001

1. Leading particulars and general data

1.1 General

This Technical Manual Part III – Overhaul and Reconditioning instructions manual is prepared to provide information about the dismantling and reassembling of the field telephones.

Chapter 2 : Describes the about the dismantling and re-assembly of the field telephone as a part of maintenance.

Chapter 3 : Describes about the Test specification and alignment procedure

Chapter 4 : Describes about specification of special components

Chapter 5 : Describes about Special Test equipment's/Jigs

Chapter 6 : Describes about Special protective measure

1.2 Abbreviations

AC	Alternative current
CBT	Common battery with DTMF dialling
CBP	Common battery with Pulse dialling
BAT MON	Battery monitoring
DTMF	Dual Tone Multiple Frequency
DC	Direct current
EMI	Electromagnetic Interference
EMC	Electromagnetic compatible
IDP	Inter Digit Pause
LED	Light Emitting Diode
LB	Local battery
LBR	Local battery with radio set
PPS	Pulse Per Second
P/T	Pulse/Tone
PCA	Printed circuit assembly
PTT	Press to Talk
RH	Relative Humidity
RV	Ringer Visual
mA	milli ampere
mV	milli volt
MTBF	Mean time between failures
MTTR	Mean time to repair
Kg	Kilogram
mV	Millivolt

2. Dismantling and Re-assembly

2.1 Removal and Replacement of PCA modules and Sub-Assemblies

The sequences to dismantle the complete telephone set along with the various PCA and sub-assembly are explained below.

Complete Telephone Set

a) Separate Handset from the telephone set.

b) Unscrew all the ten screws, mounted in the bottom assembly of the telephone, to open the top assembly of the telephone.

c) Place the top assembly and bottom assembly of the telephone unit side by side

d) Disconnect the connectors joining top and bottom assembly of the telephone unit.

In the Top Assembly

a) Dismantling of System and Volume switch knob.

b) Dismantling of System and Volume switch assembly.

c) Dismantling of Binding post PCA card.

d) Dismantling of line and external battery terminals

e) Dismantling of handset connector

f) Dismantling of buzzer assembly

g) Dismantling of hook switch PCA card

h) Dismantling of Main PCA card.

i) Dismantling of dialing keypad.

In the Bottom Assembly

a) Dismantling of ringer card PCA

b) Dismantling of battery box unit

Note - Reassembly of the unit is done by following the reverse sequence of dismantling.

2.2 Complete Telephone Set

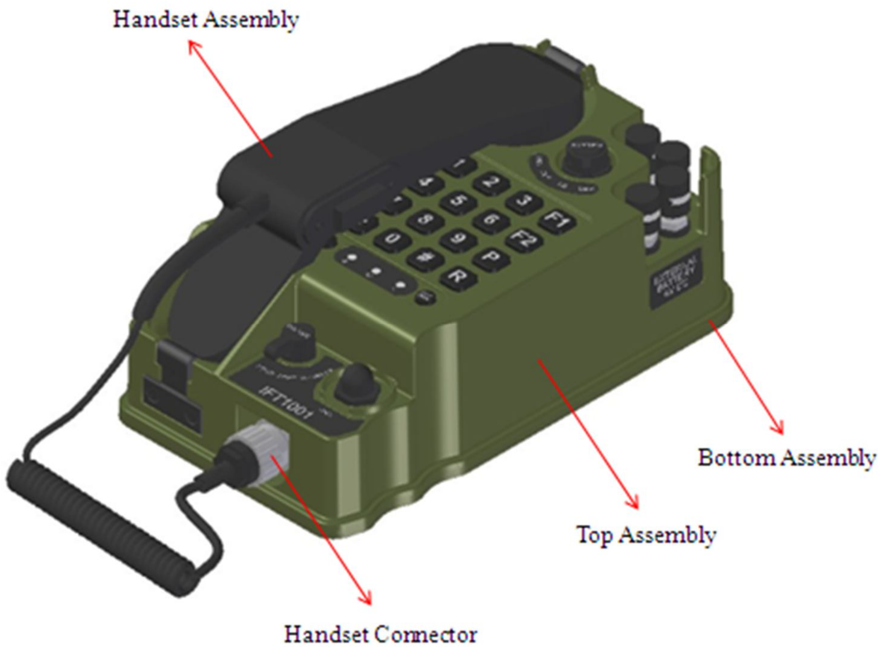


Figure 2 : Subassembly of IFT1001 telephone

2.2.1 Removal of Handset

a) Place the telephone on the flat surface.

b) Remove the handset assembly from the seat.

c) Rotate the metal cap of the handset connector with fingers and pull the connector outwards.

Note: Do not pull the plug holding the coil cord.

Fixation of Handset to Telephone

a) Insert the handset connector mounting on the coil cord of handset in to the socket part mounted in the top assembly of the telephone by observing the pin alignment of the connector.

b) Rotate clock wise, the handset connector mounting on the coil cord of handset, pushing it towards the socket until it is tighten properly as shown in Figure 3.

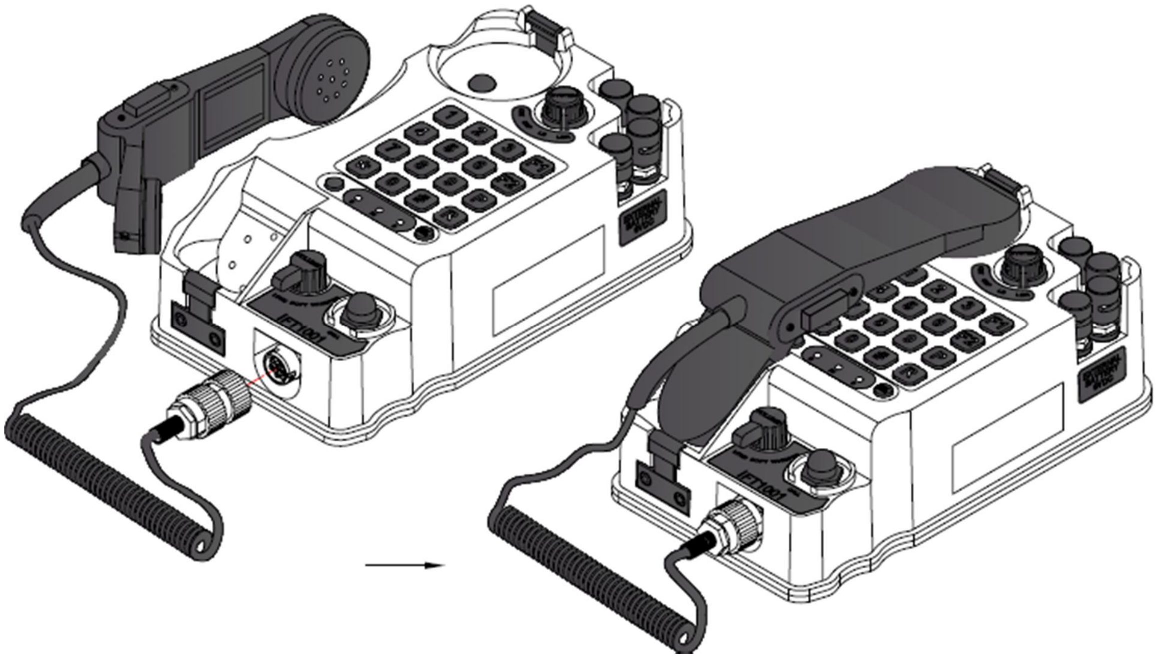


Figure 3 : Fixation of handset

2.2.2 Dismantling the Top and Bottom unit assembly

a) Place the telephone on the flat surface.

b) Remove the handset from the telephone set as describe in chapter 2.2.1.

c) Turn the telephone on the flat surface in such a way that the bottom part of the telephone is pointed upwards.

d) Unscrew completely the 10 screws marked as (5) x 10 in Figure 4.

e) Separate the Top and Bottom unit assembly of the telephone and place them side by side as shown in Figure 5.

f) Carefully remove the connector of ringer switch assembly harness (W5) and battery box assembly harness (W9) as per cable matrix in Figure 5.

g) Both the units can now physically separate from each other.

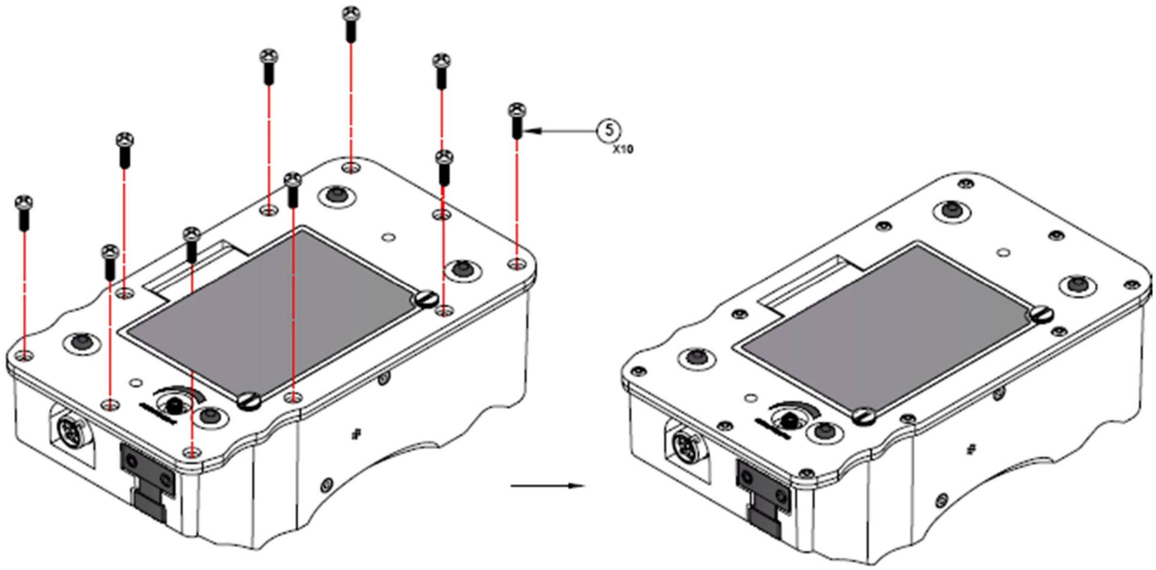
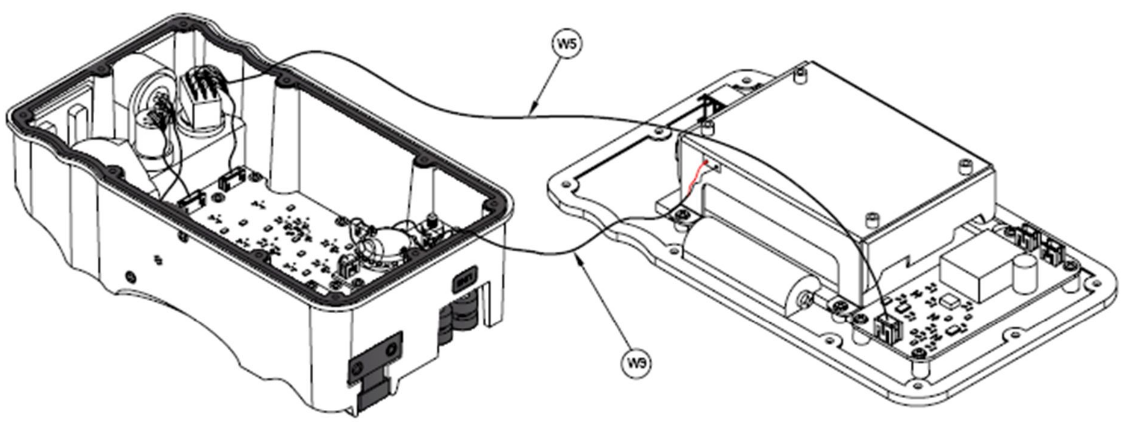


Figure 4 : Reassembly of Top and Bottom unit of telephone



Cable Matrix

Cable Ref.	Cable from	Cable to
W5	IFT1001 RINGER SWITCH ASSEMBLY	RINGER CARD (J15)
W9	IFT1001 BATTERY BOX ASSEMBLY	BINDING POST CARD (J12)

Figure 5: Top and Bottom unit of telephone after dismantling

Re-assembly of Top and Bottom unit assembly

a) Connect all the connector removed during the dismantling procedure. Observe the correct orientation of connectors during insertion.

b) Place the top unit assembly on the flat surface and mount the bottom unit assembly on the gasket mounted on the top unit assembly of the telephone.

c) Mount all the 10 screws marked as (5) x 10 in Figure 4.

d) Tighten the same using screw driver.

2.3 In the Top Assembly

2.3.1 Dismantling of System and Volume switch knob

a) Dismantle the top and bottom unit assembly of the telephone as described in the chapter 2.2.2.

b) Place the top assembly unit in the flat surface.

c) Using screwdriver, unscrew the screw (marked as 12) in the Figure 6.

d) Remove the faulty knobs i.e. System switch knob (marked as 9) or Volume switch knob (marked as 11) from the corresponding switch.

Re-assembly of System and Volume switch knob

a) Mount the required knob i.e. System switch knob (marked as 9) or Volume switch knob (marked as 11) on the corresponding switch.

b) Using screwdriver, tight the screw (marked as 12) in the Figure 6.

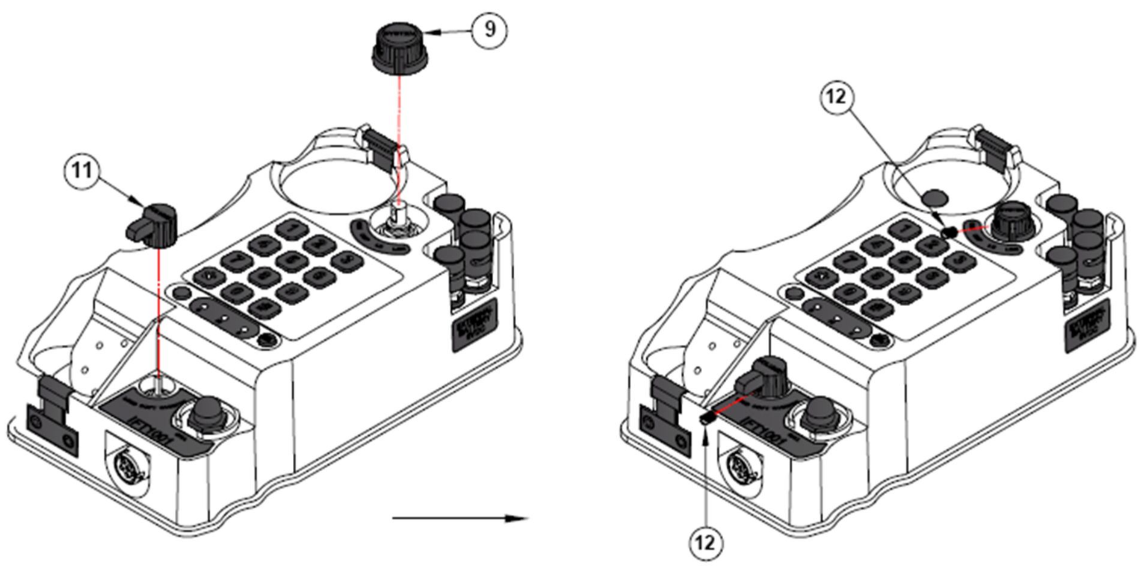


Figure 6 : Re-assembly of System and Volume switch knob

2.3.2 Dismantling of System and Volume switch assembly

a) Dismantle the System and Volume switch knob as described in the chapter 2.3.1.

b) From the top panel cover of the telephone, unscrew the nut as shown in Figure 7.

c) Remove the faulty switch i.e. System switch (W6) or Volume switch (W7) as required from the top assembly of the telephone.

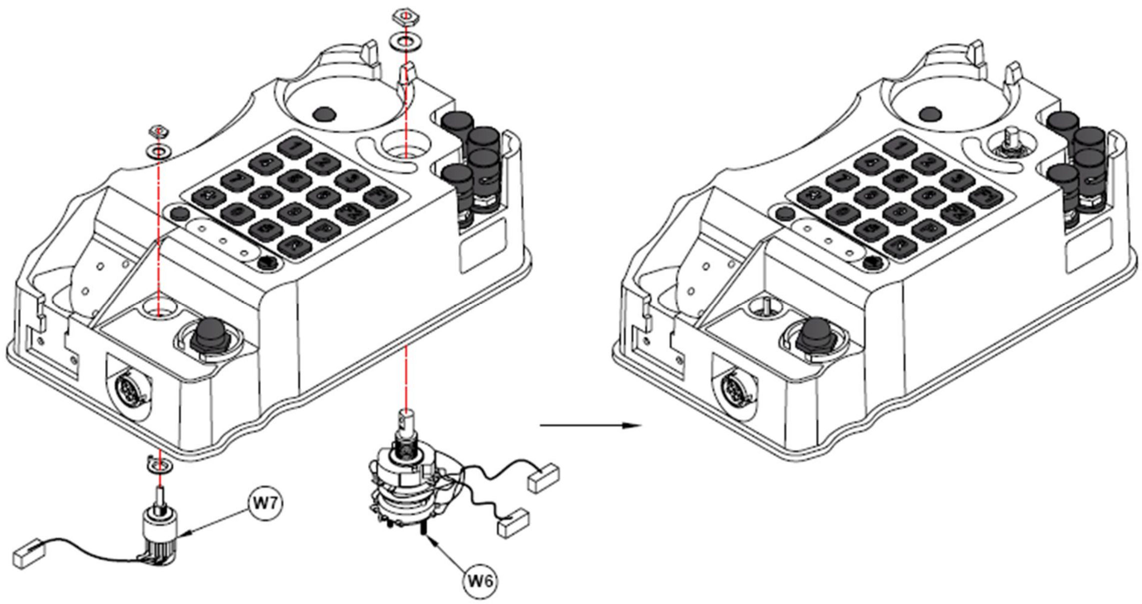


Figure 7 : Re-assembly of System and Volume switch assembly

Re-assembly of System and Volume switch assembly

a) Mount the new switch i.e. System switch (W6) or Volume switch (W7) as required from the inner side of top unit assembly of the telephone.

b) From the top panel cover of the telephone, unscrew the nut as shown in Figure 7.

c) Reassemble the System or Volume switch knob as describe in the chapter 2.3.1.

2.3.3 Dismantling of Binding post PCA card

a) Dismantle the top and bottom unit assembly of the telephone as described in the chapter 2.2.2.

b) Place the top assembly unit in the flat surface.

c) Remove the External battery connector (marked as W1) and Line terminal harness connector (marked as W2) from the Binding post PCA card.

d) Using the screw driver unscrew the screw (marked as 15) in the Figure 8.

e) Using the hexagonal nut runner, from the inner side of top assembly unit, loose the 04 nos. of nut (marked as S8) in the Figure 8.

f) Remove the plain washer (marked as S7) in the Figure 8.

g) Remove the faulty Binding post PCA card (marked as A6) mounted on the line and battery terminals in the top unit assembly of the telephone.

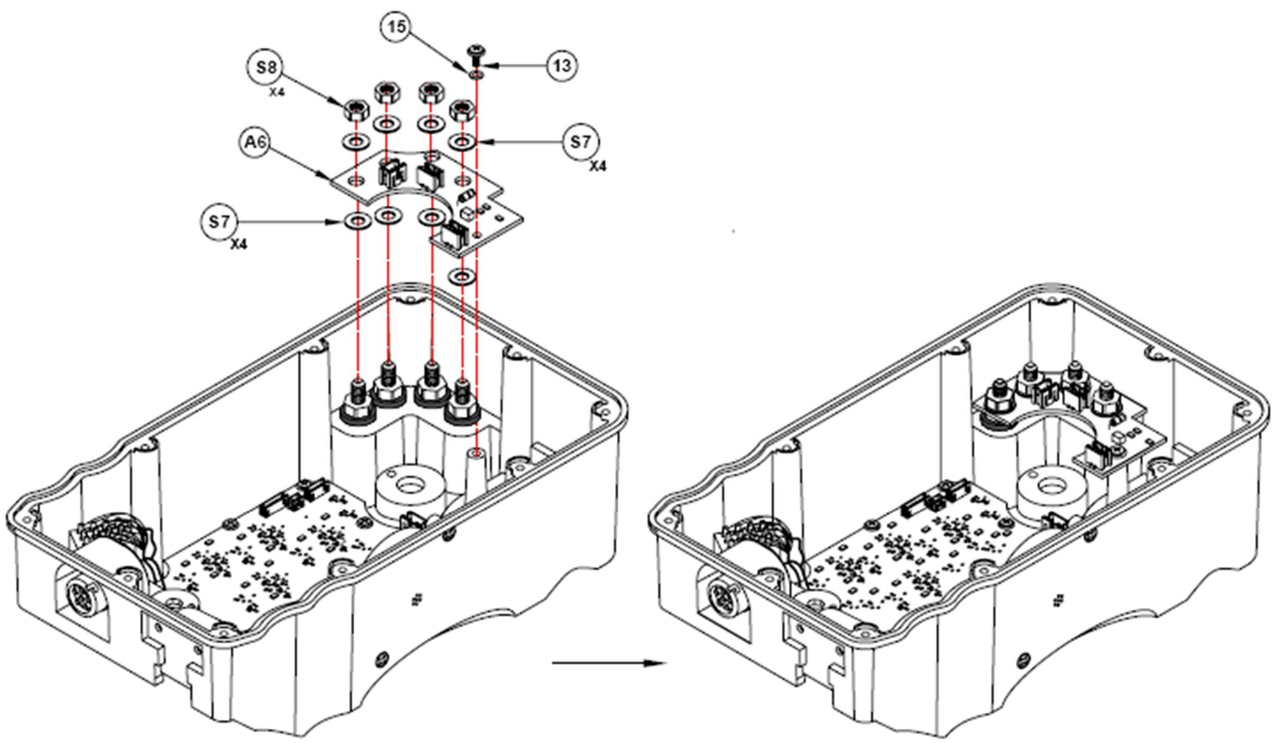


Figure 8 : Re-assembly of Binding post PCA card

Re-assembly of Binding post PCA card

a) Align the plain washer (marked as S7) on the line and battery terminals in the top unit assembly of the telephone.

b) Mount the Binding post PCA card (marked as A6) on the line and battery terminals in the top unit assembly of the telephone.

c) Mount the plain washer on the line and battery terminals over the Binding post PCA card.

d) Mount the nuts (Marked as S8) on the line and battery terminals securing the Binding post PCA card.

e) Tighten these nuts using the hexagonal nut runner.

h) Reinstall the External battery connector (marked as W1) and Line terminal harness connector (marked as W2) from the Binding post PCA card.

f) Re-assemble the Top and Bottom unit assembly of the telephone as describe in the chapter 2.2.2.

2.3.4 Dismantling of binding post (Line Terminals)

a) Dismantle the Binding post PCA as describe in the chapter 2.3.3.

b) Using the hexagonal nut driver, unscrew the 04 nos. of nut (marked as S8).

c) Remove the plain washer (marked as S7) and bush fixture for binding post (marked as S6) in the Figure 9.

d) Remove the binding post from the top panel of the telephone.

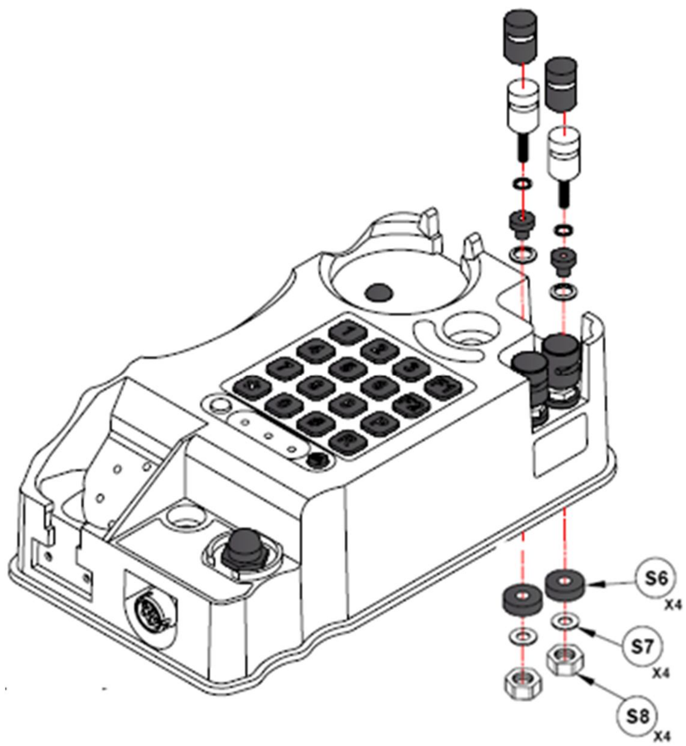


Figure 9 : Re-assembly of Binding post (Line Terminals)

Re-assembly of Binding post (Line Terminals)

a) From the top panel side of the telephone, Insert the new binding post in the holes(marked as Line).

b) Mount the bush fixture for binding post (marked as S6) and plain washer (marked as S7) on the pre mounted screw of Line Terminal.

c) Mount the hexagonal nut on the binding post (Line Terminal) and tighten the same using hexagonal nut runner.

d) Reassembly the Binding post PCA card as described in the chapter 2.3.3.

2.3.5 Dismantling of binding post (External Battery Terminals)

a) Dismantle the Binding post PCA as describe in the chapter 2.3.3.

b) Using the hexagonal nut driver, unscrew the 04 nos. of nut (marked as S8).

c) Remove the plain washer (marked as S7) and bush fixture for binding post (marked as S6) in the Figure 10.

d) Remove the External battery binding post from the top panel of the telephone.

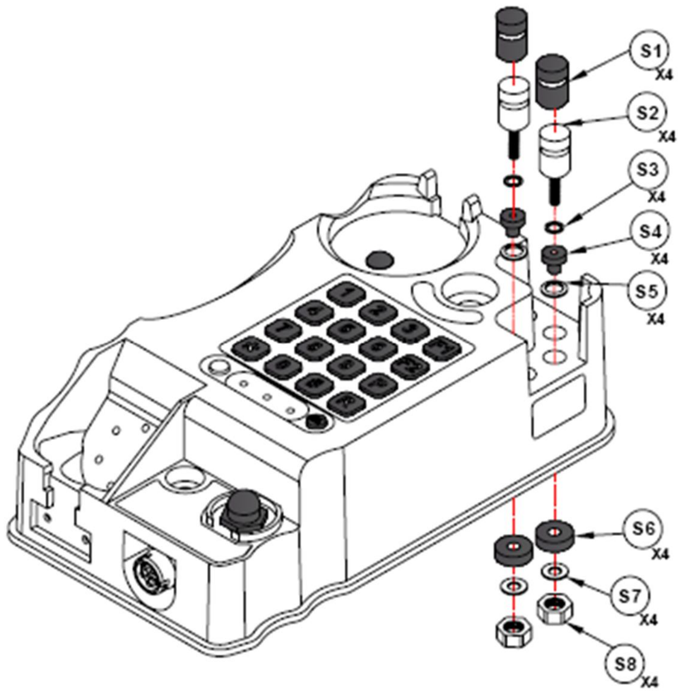


Figure 10 : Re-assembly of Binding post (Line Terminals)

Re-assembly of Binding post (Line Terminals)

a) From the top panel side of the telephone, Insert the new binding post in the holes(marked as Line).

b) Mount the bush fixture for binding post (marked as S6) and plain washer (marked as S7) on the pre mounted screw of Line Terminal.

c) Mount the hexagonal nut on the binding post (Line Terminal) and tighten the same using hexagonal nut runner.

d) Reassembly the Binding post PCA card as described in the chapter 2.3.3.

2.3.6 Dismantling of Ringer switch (MAGNETO CALL) assembly

a) Dismantle the top and bottom unit assembly of the telephone as described in the chapter 2.2.2.

b) Place the top assembly unit in the flat surface.

c) Remove the connector of Ringer switch assembly harness (marked as W5) from the main PCA of the telephone.

d) MAGNETO CALL knob contains threads. Loosen the knob of the MAGNETO CALL switch with the spanner/nut runner and pull upward.

e) Remove the locking nut of the Ringer switch.

f) Remove the Ringer switch from the Top assembly unit of the telephone.

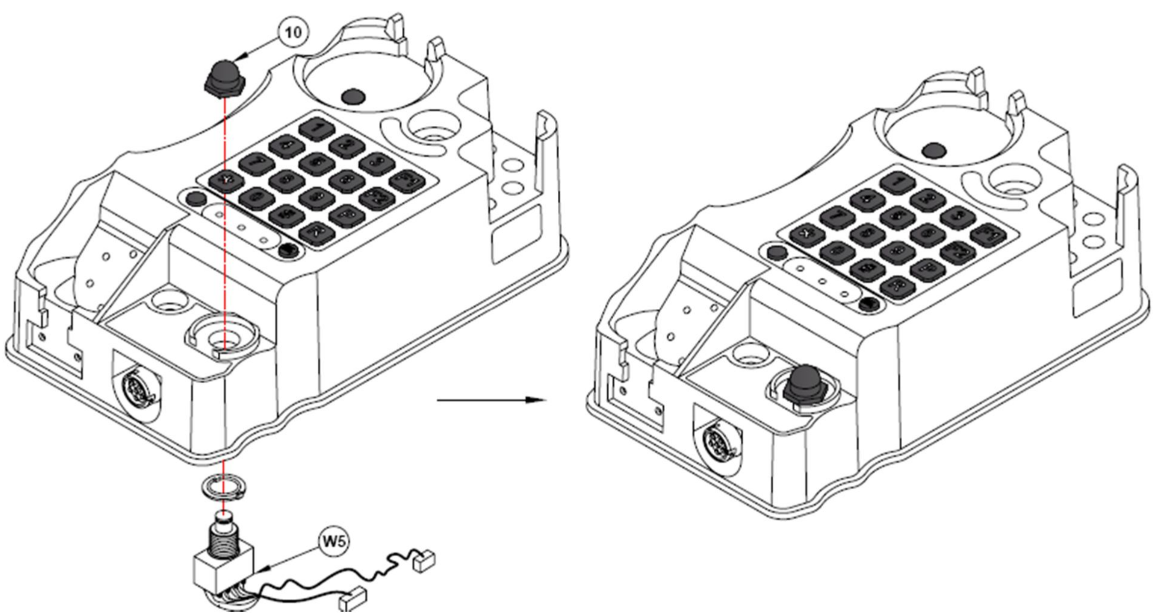


Figure 11 : Re-assembly of Ringer switch (MAGNETO CALL) assembly

Re-assembly of Ringer switch (MAGNETO CALL) assembly

a) Mount the Ringer switch in the Top assembly unit of the telephone along with the locking nut with an orientation that the locking point of the Ringer switch aligns the hole available in the Top assembly of telephone.

b) Mount the knob of the MAGNETO CALL switch from the Top panel of the telephone.

c) Tighten the same by using the spanner.

2.3.7 Dismantling of Handset connector assembly

a) Dismantle the Ringer switch (MAGNETO CALL) assembly as describe in the chapter 2.3.6.

b) Open the circular nut mounted on the Top panel of the telephone.

c) Remove the handset assembly harness connected to Main PCA.

d) Remove the handset connector assembly as shown in Figure 12.

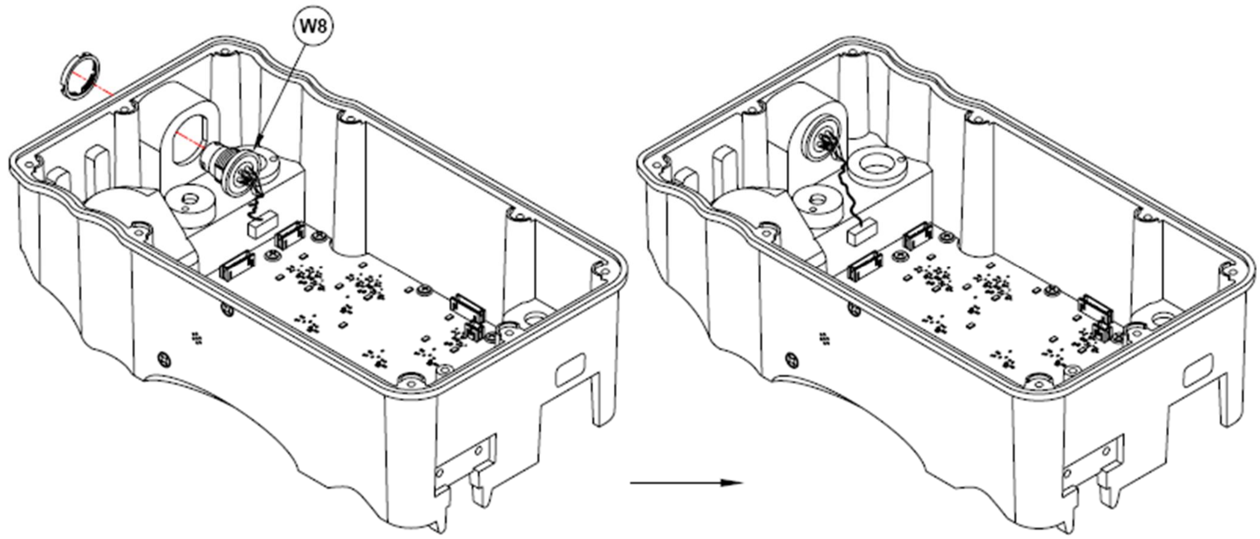


Figure 12 : Re-assembly of Handset connector

Re-assembly of Ringer switch (MAGNETO CALL) assembly

a) Mount the handset connector from the inner side of the Top assembly unit.

b) Mount the circular nut on the headset connector from the outer side of the Top assembly unit as shown in Figure 12.

c) Connect the handset connector harness (marked as W8) to the J4 connector of the main PCA.

2.3.8 Dismantling of Buzzer assembly

a) Dismantle the top and bottom unit assembly of the telephone as described in the chapter 2.2.2.

b) Place the top assembly unit in the flat surface.

c) With the screw driver, unscrew the 02 screws (marked as 12) in Figure 13.

d) Remove the buzzer assembly harness connector (marked as W4) from the J3 connector of the Main PCA.

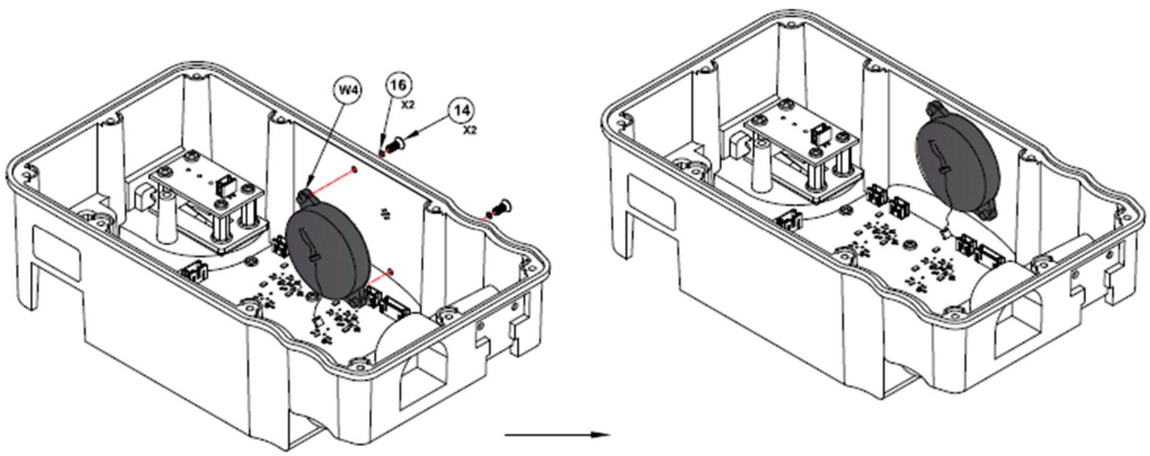


Figure 13: Re-assembly of Buzzer

Re-assembly of Buzzer assembly

a) Mount the new buzzer assembly from the inner side of the Top assembly unit.

b) Align the holes of buzzer assembly with the mounting holes on the Top assembly unit as shown in Figure 13.

c) Mount the buzzer assembly using 02 nos. of O' ring (marked as 16) and screws (marked as 14).

d) Tighten the screws on buzzer assembly using the screw driver.

2.3.9 Dismantling of Hook switch PCA assembly

a) Dismantle the top and bottom unit assembly of the telephone as described in the chapter 2.2.2.

b) Place the top assembly unit in the flat surface.

c) Using screw driver, unscrew the 04 no. of screws (marked as 13) holding the Hook switch PCA.

d) Remove the Hook switch PCA from the top assembly unit.

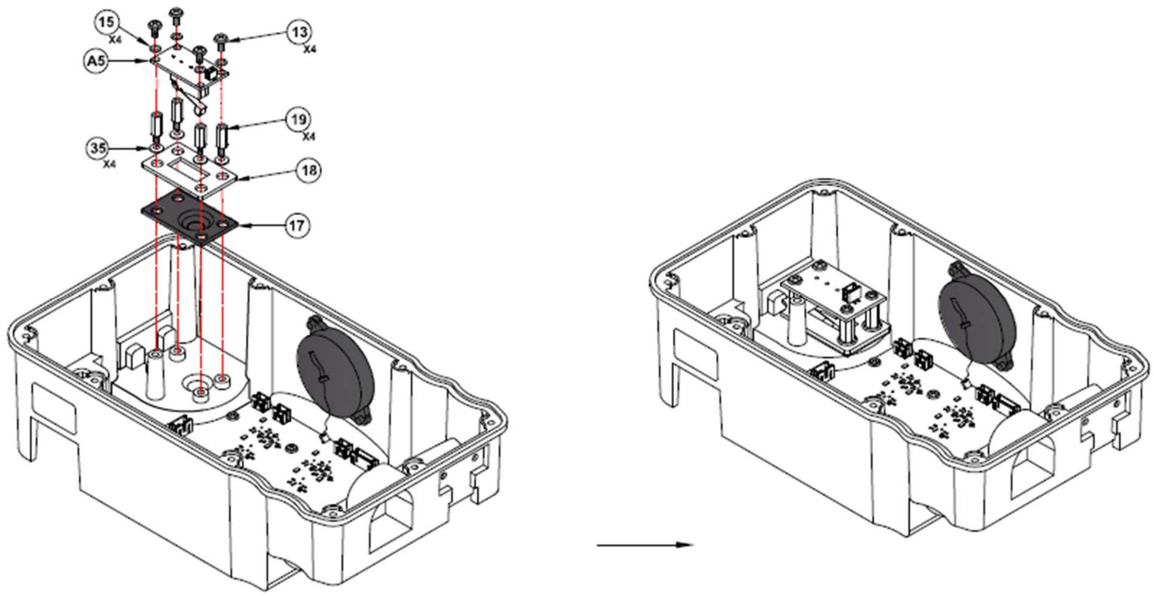


Figure 14: Re-assembly of Hook switch PCA assembly

Re-assembly of Hook switch PCA assembly

a) Mount the Hook switch PCA keeping the actuator towards hook switch cap (marked as 17) on the pillars mounted on hook switch mounting plate (marked as 18).

b) Align the 04 no. of screws on the hook switch PCA and tighten the same using the screwdriver.

c) Reassemble the Top and bottom unit assembly of the telephone as shown in the chapter 2.2.2.

2.3.10 Dismantling of Main PCA

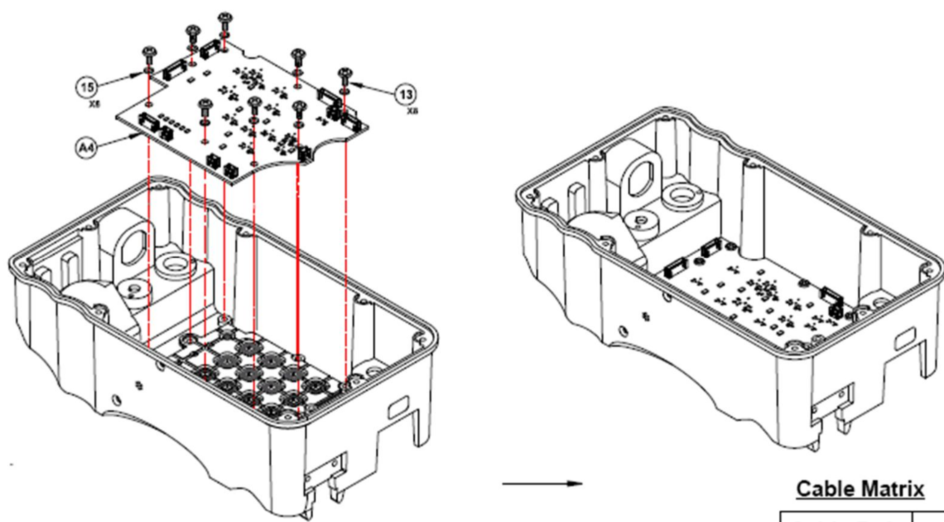
a) Dismantle the top and bottom unit assembly of the telephone as described in the chapter 2.2.2.

b) Place the top assembly unit in the flat surface.

c) Using screwdriver, unscrew the 08 nos. of screws securing the Main PCA on the Top assembly unit.

d) Remove all the cable connector mounted on the Main PCA assembly (marked as A4).

e) Remove the Main PCA from the Top assembly unit.



Cable Matrix

Cable Ref.	Cable from	Cable to
W3	HOOK SWITCH CARD (J14)	MAIN CARD (J1)
W4	IFT1001 BUZZER ASSEMBLY	MAIN CARD (J3)
W5	IFT1001 RINGER SWITCH ASSEMBLY	MAIN CARD (J6)
W7	IFT1001 VOLUME SWITCH ASSEMBLY	MAIN CARD (J5)
W8	IFT1001 HANDSET CONNECTOR ASSEMBLY	MAIN CARD (J4)

Figure 15 : Reassembly of Main PCA

Re-assembly of Main PCA

a) Mount the Main PCA on the inner side of the Top assembly unit with the connector pointed upward direction as shown in Figure 15.

b) Tighten the 08 nos. of screw with washer on the Main PCA , in order to secure the same to the Top assembly of the telephone

c) Connect all the connectors of different harness as per the cable matrix.

d) Reassemble the Top and bottom unit assembly of the telephone as shown in the chapter 2.2.2.

2.4 In the Bottom Assembly

2.4.1 Dismantling of Ringer PCA

a) Dismantle the top and bottom unit assembly of the telephone as described in the chapter 2.2.2.

b) Place the bottom assembly unit in the flat surface.

c) Remove the cable connector mounted on the Ringer PCA.

d) Using screw driver, unscrew the 04 no. of screws securing the Ringer PCA on the inner side of bottom unit assembly.

e) Remove the faulty Ringer PCA.

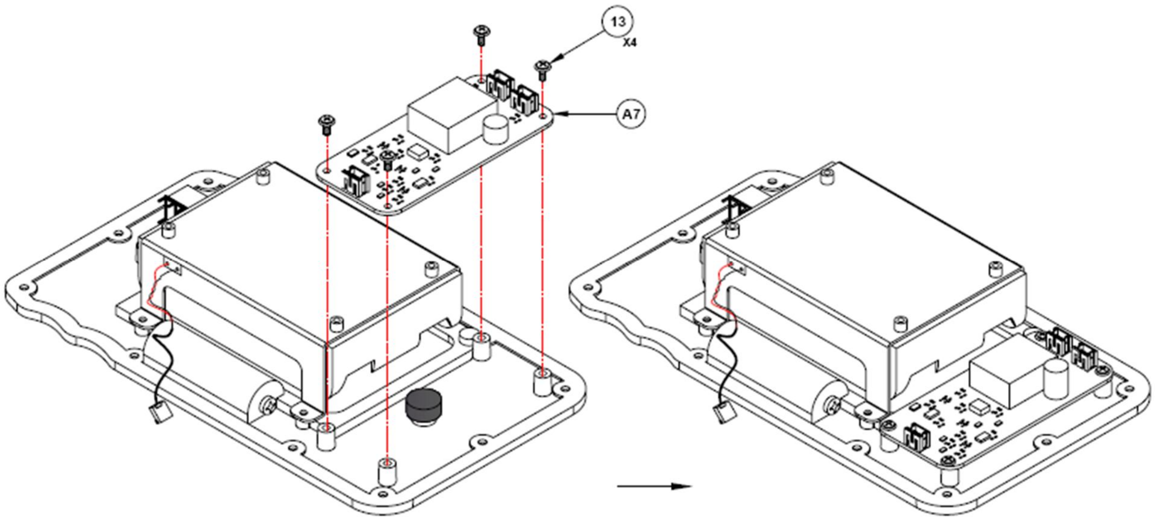


Figure 16 : Reassembly of Ringer PCA

Re-assembly of Ringer PCA

a) Mount the Ringer PCA on the inner side of the bottom assembly unit with the connector pointed upward direction as shown in Figure 16.

b) Mount the 04 no. of screws to secure the Ringer PCA on the bottom unit assembly.

c) Tighten the same by using the screw driver.

Notes

3. Test Specification and alignment procedure

Test Specification

S.No.	Specification	Range
1	Speech range	55 dB
2	Speech output (La – Lb terminals)	More than 900 mV.
3	Side tone	300 mV (minimum) below Transmit level.
4	Mode of operation: a) Magneto b) Auto Pulse/DTMF	 7500 Ohms (max.) line loop. 800 Ohms (max.) line loop (48V exchange line).
5	a) Internal battery (3 X 1.5 V 'D' type dry battery) - Voltage - Maximum current drain - Idle current - Maximum speech current b) External Battery voltage	4.5 Volt DC 180 mA Nil 8 mA 6.0 Volt dc
6	Receiver Frequency	0.3 kHz to 3.4 Khz
7	Ringing Voltage (Magneto)	90 V (minimum) at no Load
8	a) Send efficiency b) Receive efficiency	Not less than 900 mV Not less than 350 mV
9	Signaling characteristics a) Magneto b) Auto- Pulse c) Make Break ratio	 a) 25 Hz + 2 Hz b) 10 +/- 0.5 PPS; IDP 800 +/- 10% c) 60 – 70 %
10	Insulation resistance	Better than 5 Mega Ohms at 250 V DC across body and line terminals
11	Lineman test facility	With LED
12	Battery monitoring status	Good, Poor (with LED indication)
14	MTBF MTTR	More than 100000 hours Less than 15 minutes
15	EMI / EMC	As per Mil STD 461D / E

Alignment procedure

The alignment procedure for various PCA and subassembly as provided in the chapter 23.

4. Specification of Special Components

No special components are required for the maintenance of the field telephone unit.

5. Special Test Equipment/Rigs/Jigs

List of special test Equipment/Rigs/Jigs required for overhauling and reconditioning are as follows:

Description	Manufacturer Name and Model No.	Utilization
Test jig for the Main PCA Card	Elcom Innovations IFT1001 Test Jig - FT 001	Used for checking the speech circuit, buzzer circuit, supply circuit, battery monitor, status and ring indicator circuit.
Test jig for the Ringer PCA	Elcom Innovations IFT1001 Test Jig -FT 001	Used for verification of ringer circuit PCA for generating the ringer voltage.
Test jig for the Binding Post PCA	Elcom Innovations IFT1001 Test Jig -FT 001	Used for checking the functionality.
Test jig for the Hook switch PCA	Elcom Innovations IFT1001 Test Jig - FT 001	Used for checking the functionality.
Test jig for Handset assembly	Elcom Innovations IFT1001 Test Jig - FT 001	Used for checking the handset for overall operation, Microphone, Receiver and PTT.
Test Jig for Buzzer assembly	Elcom Innovations IFT1001 Test Jig - FT 001	Used for checking the operation of buzzer assembly
Special alignment tool for Line and Battery Terminal	Elcom Innovations	Used for dismantling and reassembling of Line and Battery Terminals.
13mm Tool for tightening hexagonal nut	Elcom Innovations	Used for opening on hexagonal nut of handset connector.

6. Special Protective Measure

During the Repair and overhauling of the IFT 1001 telephone, following protective measure to be observed.

a) Always align the orientation of the connector with their mating connector before insertion.

b) In order to dismantle and reassembly of any subassembly or PCA, follow the instruction given in the chapter 2.

c) While working on the Top assembly unit, care should be taken to avoid extra pressure on the same.

d) Each of the cable harness is having the marking, always mount the correct connector as per cable matrix given in this document.

7. Document History

Rev.	Change	Date	Author	Approved by	
A	New Document	09/10/2013	AA/AN	RN	

Table 1 : Document History